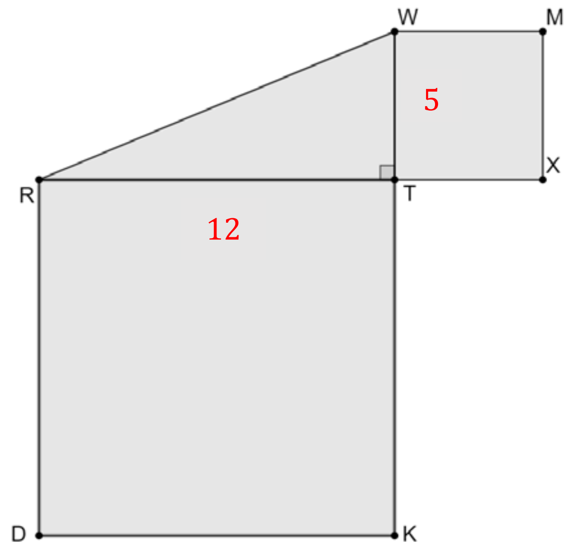
Geometry
Unit 7
Lesson 1 Homework

Name_Answer Key_________
Date____________________
Period__________________

A diagram is shown where $RT = 12$ and $WT = 5$.



Complete the statements.

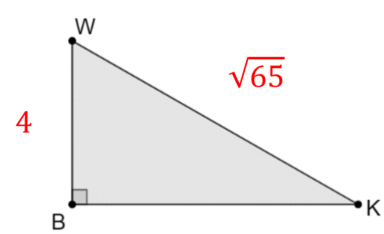
1. To show the Pythagorean theorem, a square that is drawn on the hypotenuse of $\triangle TWR$ will have an area equal to:

- o $WT + RT$.
- o $(WT + RT)^2$.
- o the sum of the area of square $WMXT$ and the area of square $RTKD$.

2. The length of each side of the square that is drawn with side $\overline{RW}$ is 13.

$$5^2 + 12^2 = c^2$$
$$c = 13$$

3. $\triangle WKB$ is shown where $WK = \sqrt{65}$ and $WB = 4$. Find the exact length of $\overline{BK}$.



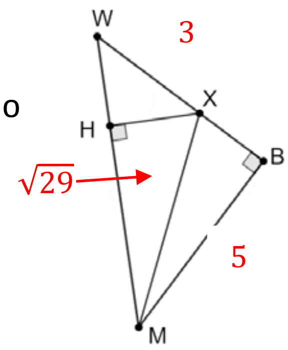
$4^2 + b^2 = \sqrt{65}^2$ $\quad b = 7 \quad$ $\boxed{BK = 7}$

4. A right triangle has side lengths of $\sqrt{35}$ and $\sqrt{52}$. Find the exact length of the hypotenuse.

$$\sqrt{35}^2 + \sqrt{52}^2 = c^2$$

$$c = \sqrt{87}$$

5. Triangle WBM is shown where $m\angle WBM = 90°$ and $m\angle XHM = 90°$. $MX = \sqrt{29}$, $WX = 3$, and $BM = 5$.
Determine the length of $\overline{WM}$. If necessary, round your answer to the nearest thousandth.



$XB: a^2 + 5^2 = \sqrt{29}^2$

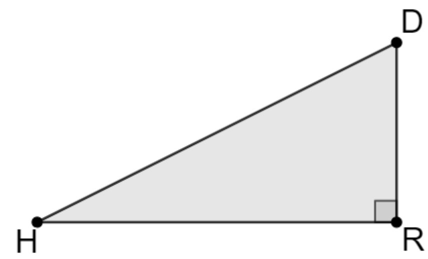
$a = 2$

$WB: 3 + 2 = 5$

$WM: 5^2 + 5^2 = c^2$

$c = \boxed{WM = 7.071}$

$\triangle DRH$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table.



	Length of $\overline{DR}$	Length of $\overline{HR}$	Length of $\overline{DH}$
6.	28	15	31.765
7.	36	77	85
8.	17.493	$\sqrt{18}$	18
9.	33	56	65
10.	5	$\sqrt{7}$	$\sqrt{32}$

6) $28^2 + 15^2 = c^2$
$c = 31.765$

7) $36^2 + b^2 = 85^2$
$b = 77$

8) $a^2 + \sqrt{18}^2 = 18^2$
$a = 17.493$

9) $33^2 + b^2 = 65^2$
$b = 56$

10) $a^2 + \sqrt{7}^2 = \sqrt{32}^2$
$a = 5$